

CHAPTER 16

Understanding Exposure in Physical Trading

How price risk builds up in a physical cargo — and how traders measure, decompose and hedge it.



WHY IT MATTERS

You cannot hedge what you cannot measure

Every physical cargo carries a web of price exposures: flat price risk, calendar spread risk, basis risk, and crack risk. A trader who only tracks flat price while ignoring the rest may believe they are hedged — and discover in settlement that they were not. Exposure management starts before the hedge, not after.

4 layers

of price risk in any physical cargo: flat price, structure, basis, and crack — each settles at a different time and on a different benchmark

Flat price is only the beginning

The cargo may also carry a month spread, a Dated-vs-ICE differential, and a crack — all of which can move against you independently.

Exposure changes daily

Pricing quotation periods advance each day, so the risk in a book evolves even if no new deals are done.

FOUNDATIONS

Physical vs Priced Exposure

Exposure does not begin when oil is loaded — it begins when the pricing period starts, and it resolves only when every quote is fixed.

PHYSICAL EXPOSURE

Cargo bought, not yet priced

You are contracted to lift the cargo. The physical commitment is locked in. But the quotation period for pricing has not yet started — so you do not yet know the final number, and the market risk has not fully materialised.

Example: You buy a cargo of Crude pricing on the average of next week's Platts quotes. The cargo is committed — but the price is still open.

PRICED EXPOSURE

Quotation period active

The pricing window is open. Each day that passes, Platts publishes a quote that feeds into the average. Each un-fixed quote is an active price risk. As quotes are fixed, the open exposure shrinks until the cargo is fully priced.

Example: 10 of 22 pricing days have already settled. The remaining 12 quotes are still live — that is your current priced exposure in lots.

As each quote is fixed, it is "priced out" — the exposure on that day disappears. A fully priced cargo carries zero market risk.

TAXONOMY

The four types of price exposure

Physical oil trading generates price risk across four dimensions. Each is independent — moves in one do not eliminate risk in another.

01

Flat Price

ICE BRENT / NYMEX WTI

The outright level of the benchmark price. Long if you bought at a floating price; short if you sold fixed. The largest single source of P&L volatility for most players.

02

Structure

CALENDAR SPREADS (M1 VS M2)

The difference between two futures contract months. A cargo priced in one month versus a hedge booked in another creates calendar spread risk. Moves with inventory and carry.

03

Basis / Differential

DATED VS ICE (DFLS)

The gap between a physical assessment (e.g. Dated Brent at 16:30) and the exchange price (ICE at 19:30). Changes daily and can diverge sharply around roll periods.

04

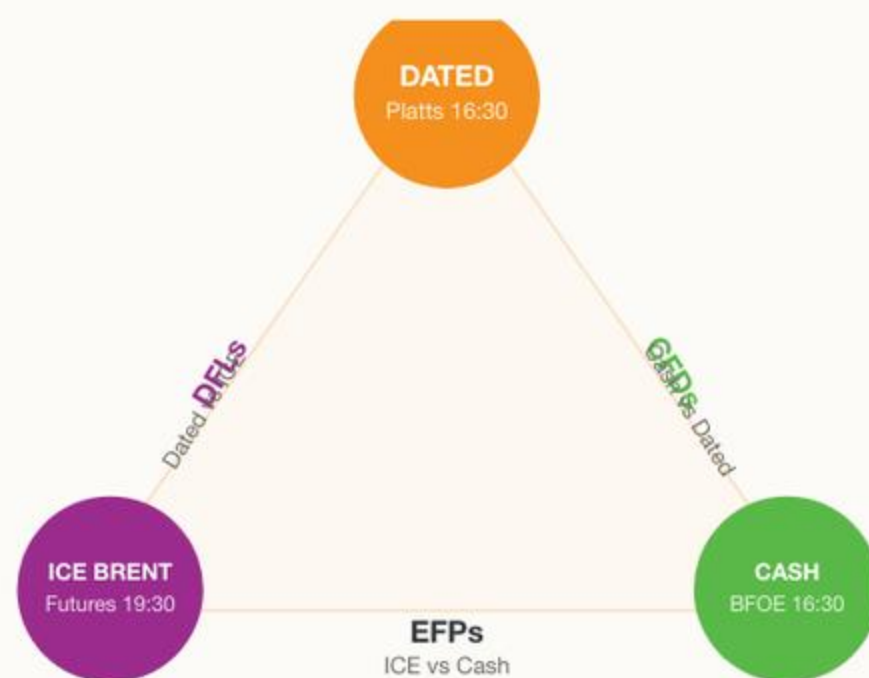
Crack / Product Spread

PRODUCT VS CRUDE

For refiners and product traders: the margin between the product price and the crude feedstock. Gasoil crack or naphtha crack risk is carried from purchase through to sale.

The Dated Triangle

Brent pricing has three distinct nodes, each with its own assessment. Physical trades move exposure between them — and each connection generates a different type of risk.



Dated Brent — the physical anchor

Platts assessment published at 16:30 London. This is what physical cargoes (crude, CIF product) typically price against. It is the real-world benchmark.

ICE Brent Futures — the liquid hedge

Exchange-traded contract settling at 19:30. The most liquid oil futures contract in the world. Hedges booked here settle on a different price and at a different time than Dated.

Cash (BFOE) — the forward physical

Forward physical lots in 100,000-bbl partials, priced at 16:30. Six hundred lots make one cargo. Bridges the gap between the Dated assessment and futures settlement.

Each leg — **DFLs**, **CFDs**, **EFPs** — is a paper instrument that lets traders move exposure between nodes, managing the differential risk that arises between them.

POSITION MECHANICS

Long and Short: how a trade builds exposure

Every physical deal creates a directional exposure. Understanding long and short is the starting point for all exposure analysis.

LONG

Net buyer — benefits if price rises

You have bought a floating-price cargo that has not yet been re-sold or hedged. Every \$1/bbl rise in the benchmark adds to your P&L. Every \$1/bbl fall costs you.

SHORT

Net seller — benefits if price falls

You have sold a cargo on a floating price without covering the purchase. Every \$1/bbl fall in the benchmark profits you. A rising market works against you.

Exposure is measured in **lots** (1 lot = 1,000 bbls). A 500,000-bbl cargo with 22 pricing days open = **500 lots** of flat price exposure remaining.

How exposure builds: cargo lifecycle

1

Deal struck

Buy 500,000 bbls crude, pricing on monthly average Dated Brent. Position is LONG 500 lots flat price.

2

Pricing period opens

Each day Platts publishes a quote. 10 of 22 quotes fixed → $10/22 \times 500 = 227$ lots priced out. Remaining: 273 lots open.

3

Hedge is placed

You sell 500 ICE Brent futures. Flat price risk is now flat. But structure and basis exposure remain.

4

Pricing period closes

All 22 quotes are fixed. Flat price exposure = 0. Futures position is offset at expiry. P&L is locked.

Flat Price Exposure

A simple buy/sell structure shows how flat price exposure arises — and why hedging it requires selling the same number of futures lots as the unpriced quota.

THE TRADE	
Buy 500,000 bbls crude	Floating price
Sell 500,000 bbls crude	Fixed \$82.00/bbl
Pricing basis	Monthly avg Dated Brent

Exposure analysis

Purchase side	LONG 500 lots — floating price exposure
Sale side	FLAT — fixed price, no market risk
Net position	LONG 500 lots flat price (Dated Brent)
P&L sensitivity	+\$500K per \$1/bbl move in Dated Brent

Hedging the flat price risk

- ### Step 1 — Sell ICE Brent futures

Sell 500 lots of the relevant ICE contract month. This neutralises the flat price. If Dated Brent falls \$3/bbl, the physical buy loses \$1.5M — but the futures gain \$1.5M.
- ### Step 2 — Recognise residual exposure

The futures hedge is against ICE, not Dated. The DFL (Dated vs ICE spread) remains open. A widening DFL works against a buyer who is long Dated, short ICE.
- ### Step 3 — Hedge the DFL separately

Sell a DFL to lock in the Dated–ICE differential. Now the position is flat on both flat price and the Dated–ICE basis. Only structure (calendar spread) risk may remain.

Key insight: Hedging is always multi-step. Lock flat price first; then address each remaining layer — DFL, CFD, structure.

Structure Exposure

Even a perfectly flat-price-hedged book can carry significant structure risk. Structure exposure arises when the pricing month of a physical trade differs from the futures contract used to hedge it.

Month/Month structure example

Item	Nov	Dec
Physical buy price	MOV Nov avg	—
Futures hedge	Sell Nov ICE	—
Physical sell price	—	MOV Dec avg
Futures hedge needed	—	Buy Dec ICE
Resulting structure	Nov priced	Dec priced

The trader is **long Nov / short Dec**. This is a Nov–Dec calendar spread position. If Nov weakens relative to Dec (backwardation narrows), the trade loses money — even if flat price is unchanged.

How structure risk builds

- ### Mismatched months create spread risk

Any trade where the purchase month $\neq$ sale month generates a calendar spread position. The P&L depends on how the spread moves — not on the level of flat price.
- ### Hedge the structure with a dated roll

Buy the BAL NOV vs CAL DEC Dated Roll (or equivalent futures spread) to offset the month mismatch. The spread trade neutralises the residual structure risk.
- ### DTD roll for day-to-day structure

Fine-grained structure risk — where the cargo prices on different days relative to the hedge — is managed via the DTD (day-to-day) roll, also known as a dated roll.
- Rule of thumb:** after hedging flat price and the DFL, whatever month mismatch remains in the futures book is your structure exposure. Hedge it with the same-month spread.

DATED VS ICE

Basis Exposure and DFLs

A DFL (Dated Front Line) is the OTC swap that captures the difference between the Dated Brent assessment and the prompt ICE Futures contract. It is the key instrument for managing basis exposure.

WHAT IS A DFL?

Dated Front Line = **Dated Brent (16:30)** minus **ICE Brent Futures (19:30)**. It is an illiquid OTC instrument that trades as a spread. A long DFL means you benefit if Dated strengthens relative to ICE; a short DFL means you benefit if it weakens.

Why the basis matters

Different time: Dated at 16:30, ICE at 19:30 — 3 hrs of market movement embedded in the spread

Different liquidity: ICE futures are highly liquid; Dated is an OTC assessment — the two can diverge sharply

Roll risk: Near roll periods the DFL swings as the prompt month changes — classic basis trap for the unwary

DFL mechanics — worked example

SCENARIO

- Physical: Buy cargo: **Pricing vs Dated Brent (16:30) — LONG Dated**
- Hedge: Sell ICE futures: **SHORT ICE Brent (19:30) — flat price hedged**
- Residual exposure: **LONG DFL = Dated – ICE spread**
- DFL hedge: **Sell DFL to counterparty — basis risk neutralised**
- Final position: **Flat on flat price AND basis — only structure remains**

DFLs typically face a counterparty directly (OTC). Platts publishes reference prints but these will not align exactly with where Dated and Futures settle — adding a further layer of risk in the print itself.

EFPs — Bridging ICE and Cash

Exchange for Physical (EFP) is the mechanism traders use to convert a futures position into a physical one — or vice versa — at the agreed ICE–Cash spread.

WHAT IS AN EFP?

An OTC swap valued at the difference between ICE Brent Futures and Cash (BFOE) prices. A **long EFP** = sell ICE, buy Cash. A **short EFP** = buy ICE, sell Cash. It prices like a DFL but connects Cash to ICE rather than Dated to ICE.

September 2020 market example

Contract	Cash	ICE Nov
Nov Settlement	42.81	42.72
ICE Futures leg	Nov Cash: 42.81	Nov Futures: –42.72
EFP spread	+0.09	Printed at Cash–ICE

The EFP print (0.09) matches the Cash–ICE spread on that date. By trading the EFP, the trader has effectively moved their hedge from the exchange into the physical market.

When to use an EFP

Converting a futures hedge to physical

After hedging flat price with ICE futures, the trader can turn that futures position into a Cash (BFOE) position via an EFP — useful if the physical desk wants to take or make delivery.

Moving ICE @ MOPs hedges

Many physical contracts price "ICE @ MOPs" (ICE at MOPs timing). The EFP allows a trader to move between Cash pricing and ICE pricing, locking in the differential at a specific level.

Avoiding delivery on futures

A trader long ICE futures approaching expiry who does not want physical delivery can use an EFP to exit the futures and enter a forward physical position with a counterparty instead.

EFP vs DFL: DFLs link *Dated* to *ICE*. EFPs link *Cash (BFOE)* to *ICE*. Both are OTC and illiquid relative to outright futures, but essential for moving exposure around the Dated Triangle.

RISK MANAGEMENT IN PRACTICE

Measuring and Netting Exposure

The exposure report — updated daily — nets physical cargoes, hedges and paper trades into a single risk picture across all four dimensions.

Daily exposure report — simplified example

Trade / Position	FP lots	DFL lots	M1/M2	Crack
Buy ESPO cargo (Oct)	+500	+500	+500 Oct	—
Sell GoA cargo (Nov)	-500	-500	-500 Nov	—
ICE futures hedge	-500	—	+500 Nov	—
DFL hedge sold	—	-500	—	—
Gasoil crack buy	—	—	—	+200
NET POSITION	0	0	Oct/Nov spread	+200

Flat price is flat, DFL is flat — but an Oct/Nov calendar spread and a gasoil crack remain open. These must be managed separately.

Key exposure management principles

Net across the book, not trade by trade

Multiple cargoes may offset each other. A desk long 300 lots in one trade and short 300 in another has zero net FP — but each leg still has DFL and structure risk.

Update exposure daily

Quotation periods advance every day. 5 days settled overnight → 5 lots less exposure this morning. Risk profile changes even without new deals.

Set and monitor limits

Every dimension — FP, DFL, structure, crack — should have a risk limit. Breaches trigger mandatory hedge action, not a conversation.

Watch the correlation

Exposures that appear to cancel on paper can diverge in stress. Dated and ICE separated sharply in COVID March 2020 — an assumed-flat DFL was not flat.

KEY TAKEAWAYS

Know your exposure before you hedge

Exposure is multi-layered

Every physical cargo carries flat price, basis (DFL), structure (calendar spread), and potentially crack risk. Each layer settles independently and must be hedged separately.

The Dated Triangle is the map

Dated, ICE Futures, and Cash are three distinct pricing nodes. DFLs, CFDs, and EFPs are the bridges. Moving exposure around the triangle is how traders manage basis and delivery risk.

Exposure changes daily

Quotation periods advance every business day. Physical exposure shrinks as each price is fixed. A good trader updates their view of the book every morning — before the market opens.

The professional standard: a trader should be able to state, at any moment, their net exposure in every dimension — lots of flat price, lots of DFL, calendar spread by month, and crack exposure. If they cannot, they are flying blind.